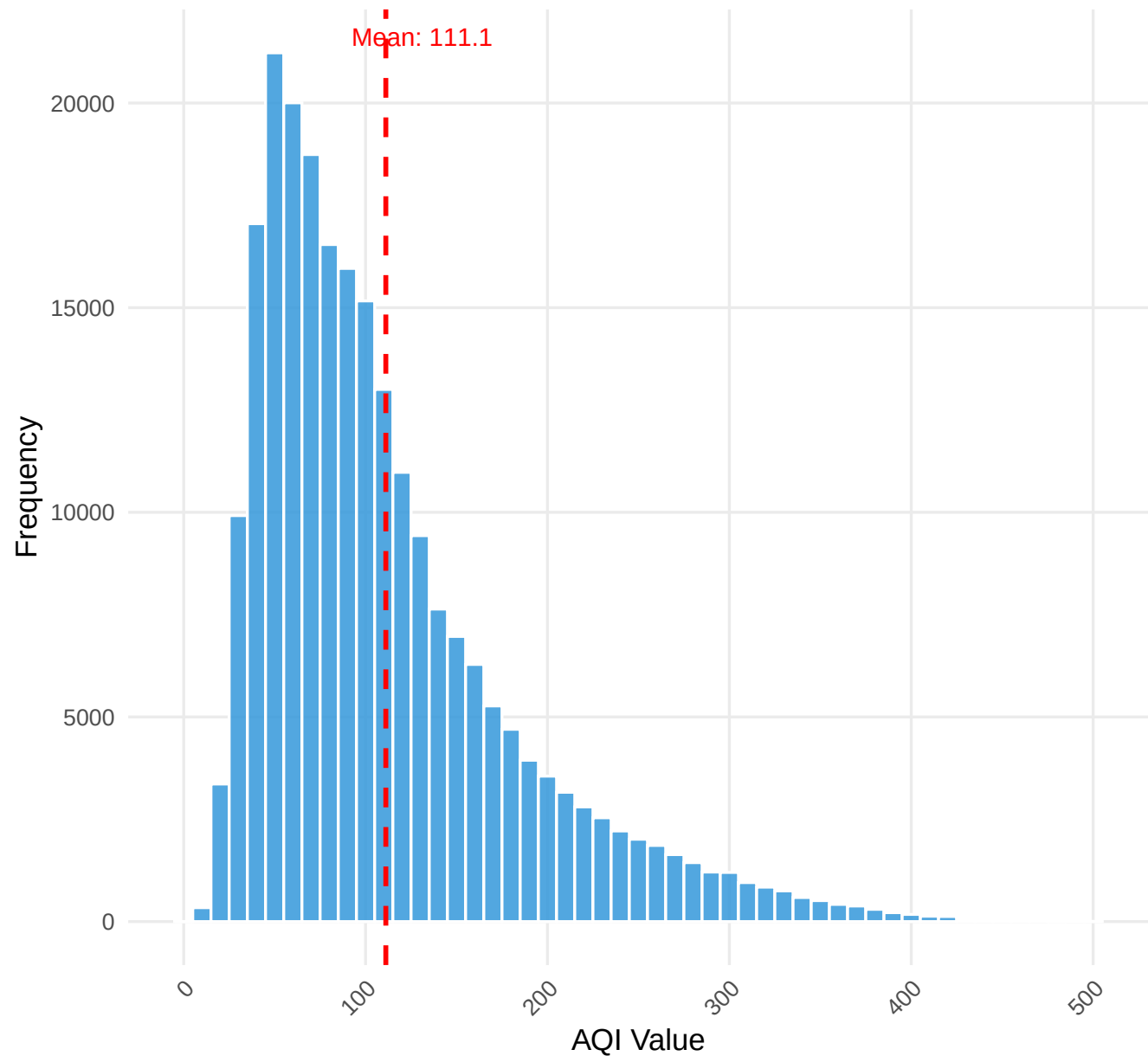


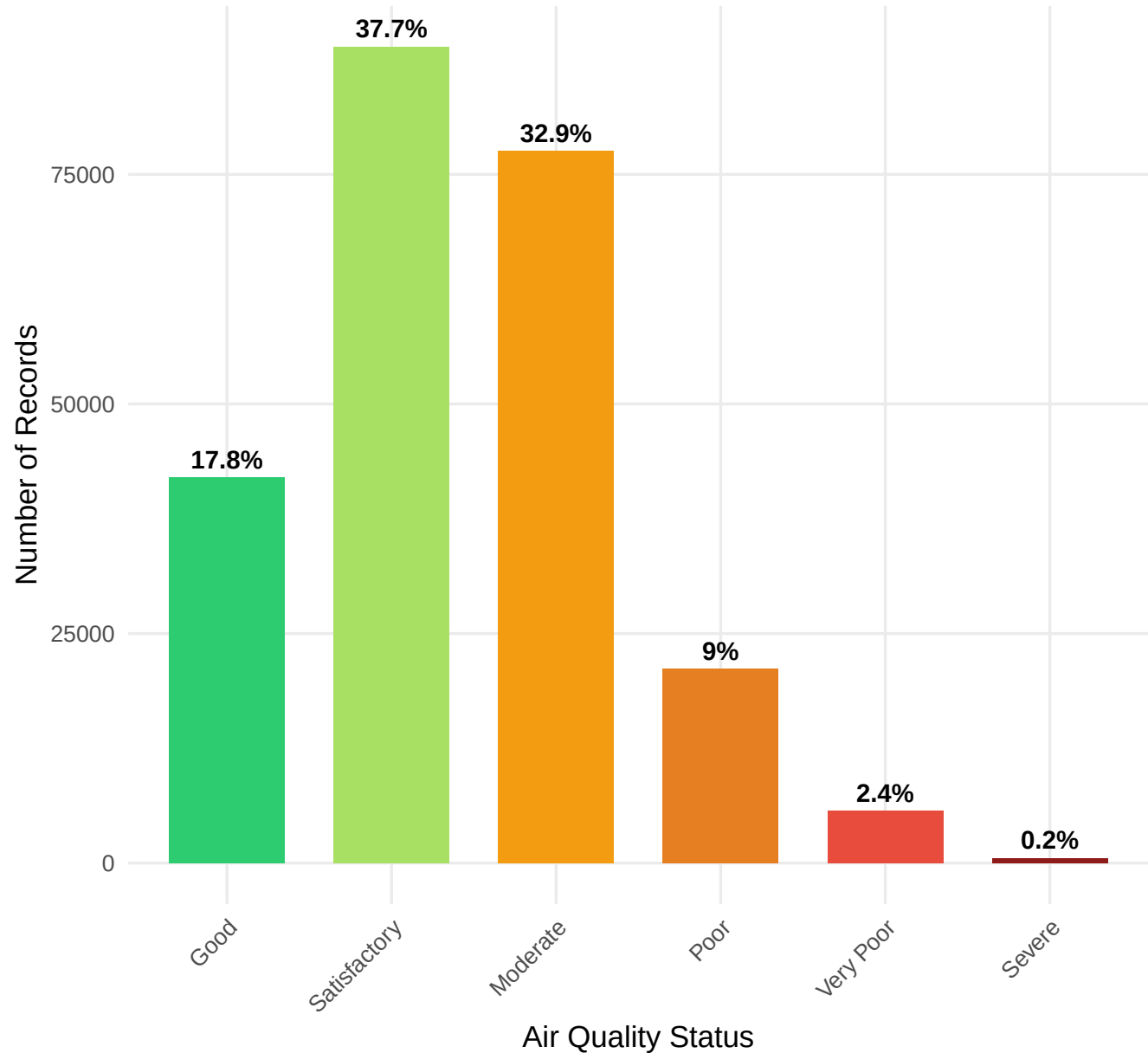
Distribution of AQI Values Across India

Red dashed line represents the mean AQI



AQI Category Distribution

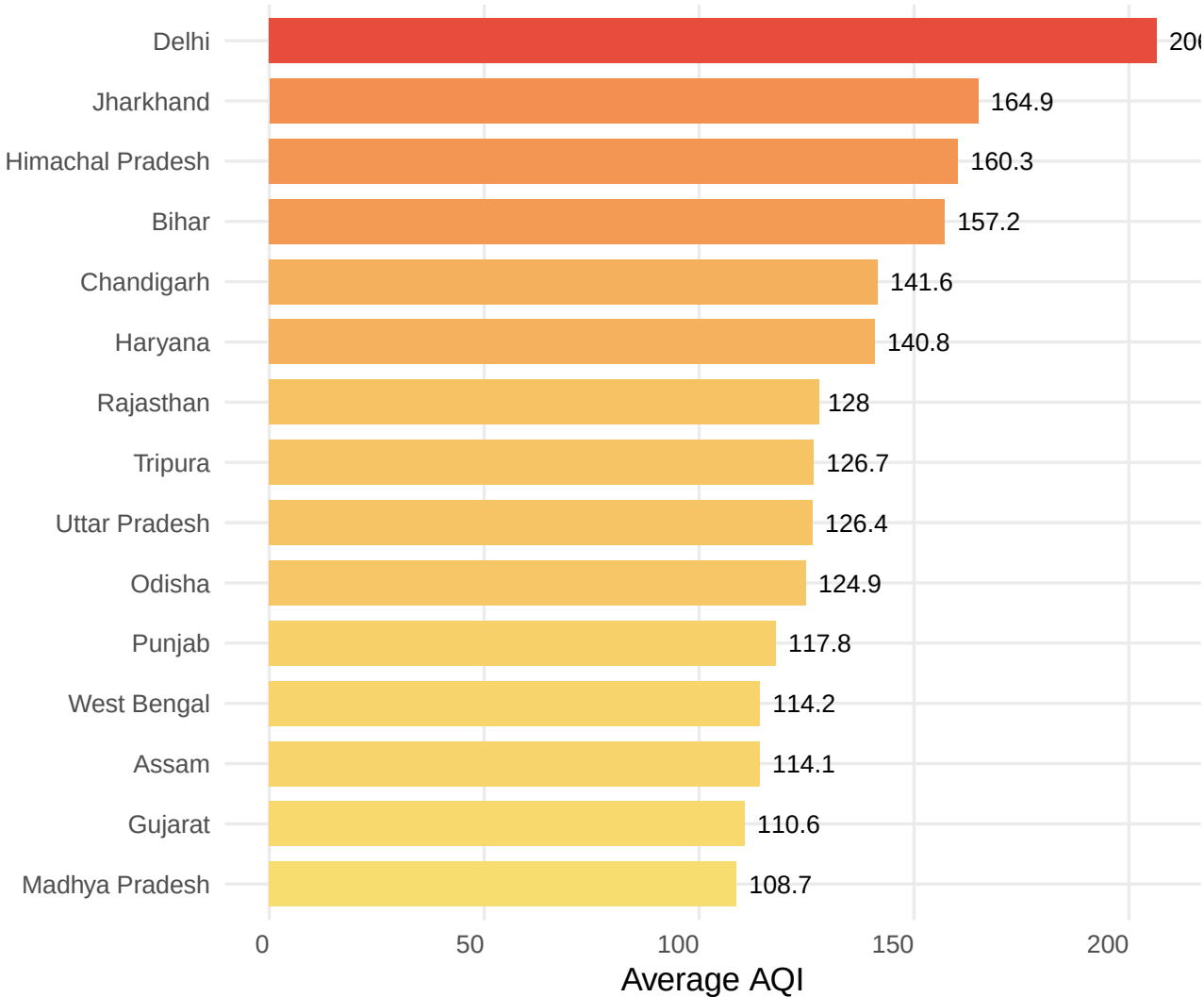
Count of records per air quality category



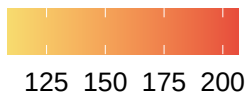
Top 15 Most Polluted States

Ranked by Average AQI

State

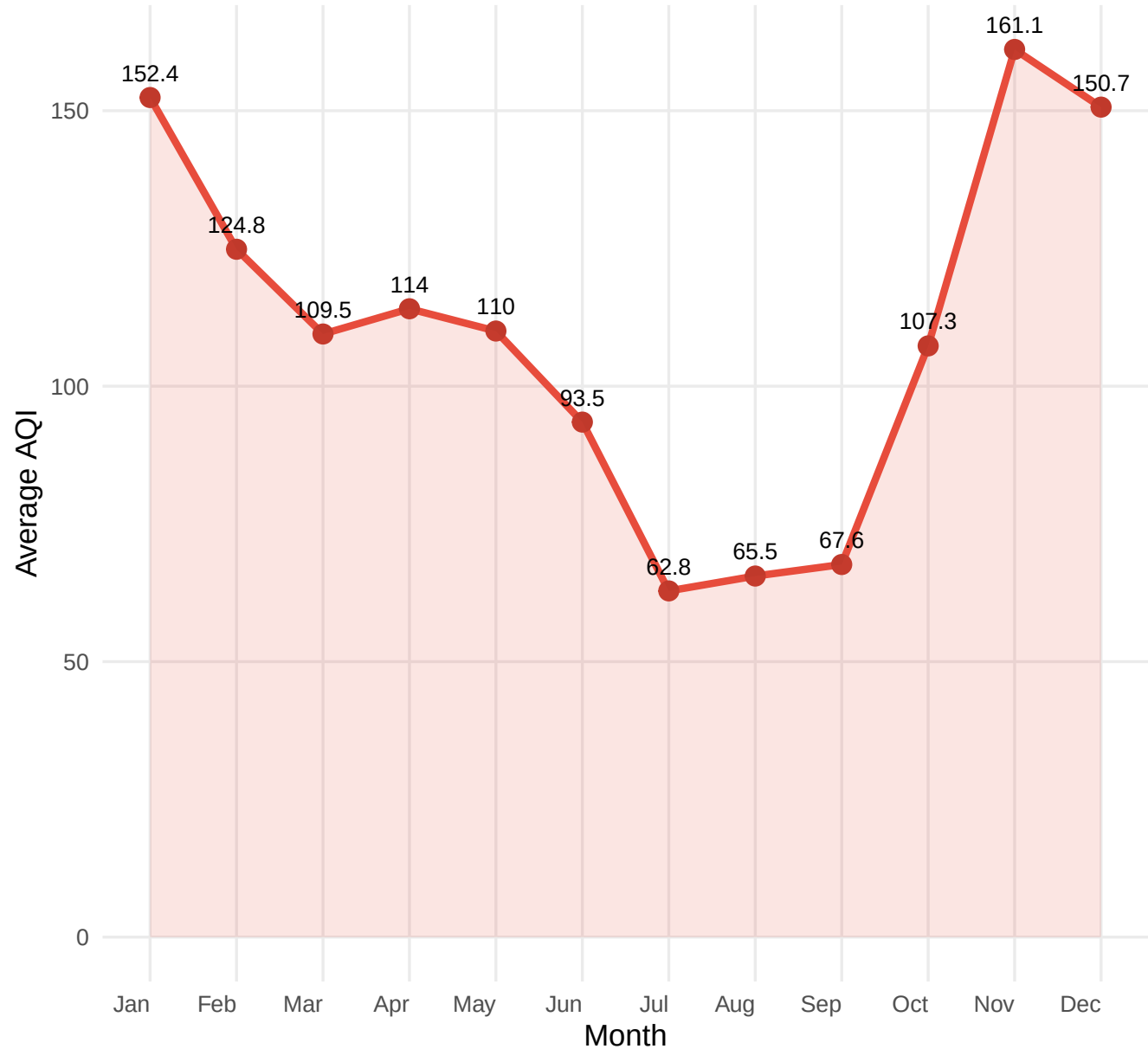


Avg AQI



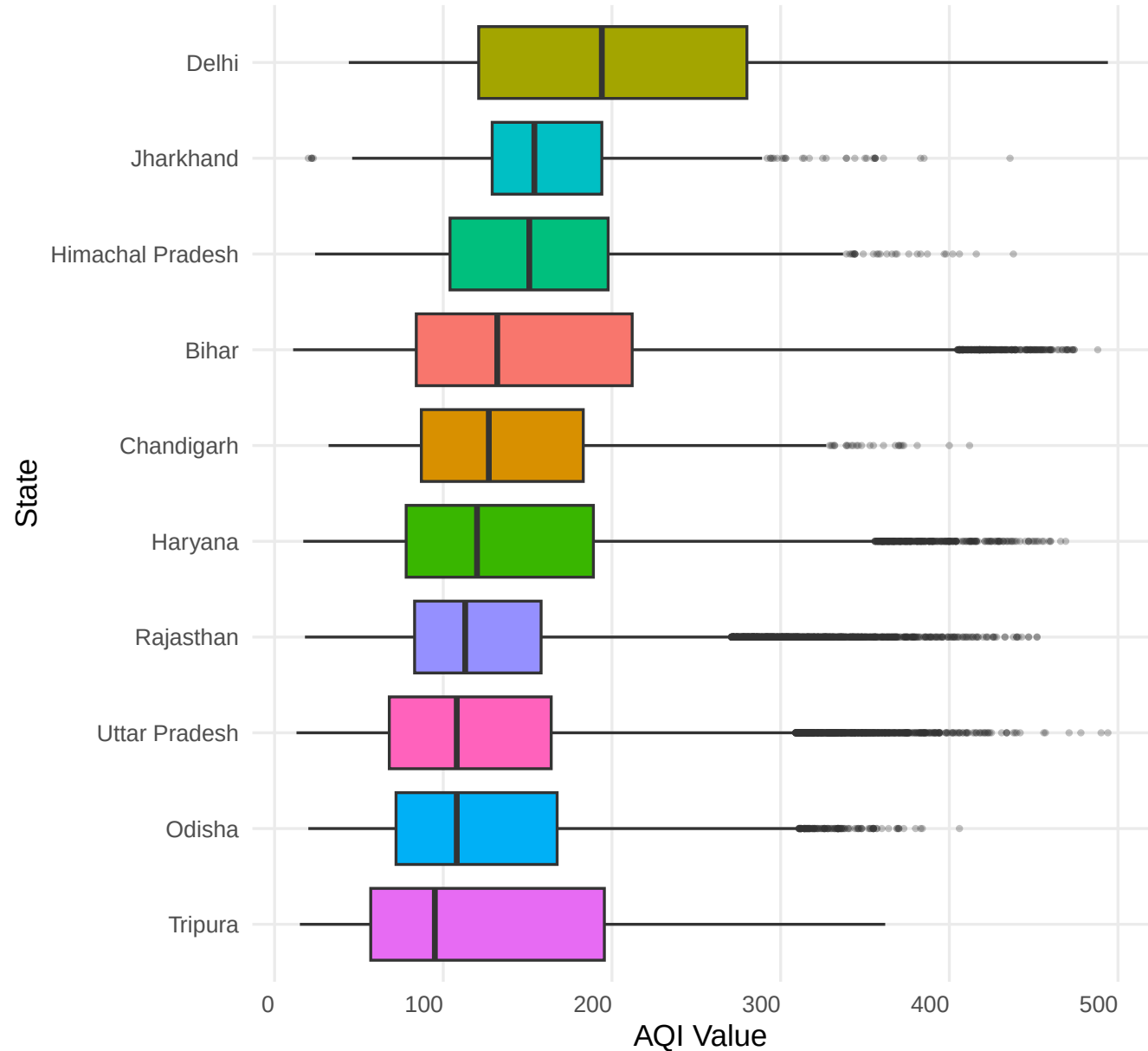
Monthly AQI Trend

Average AQI across all cities by month



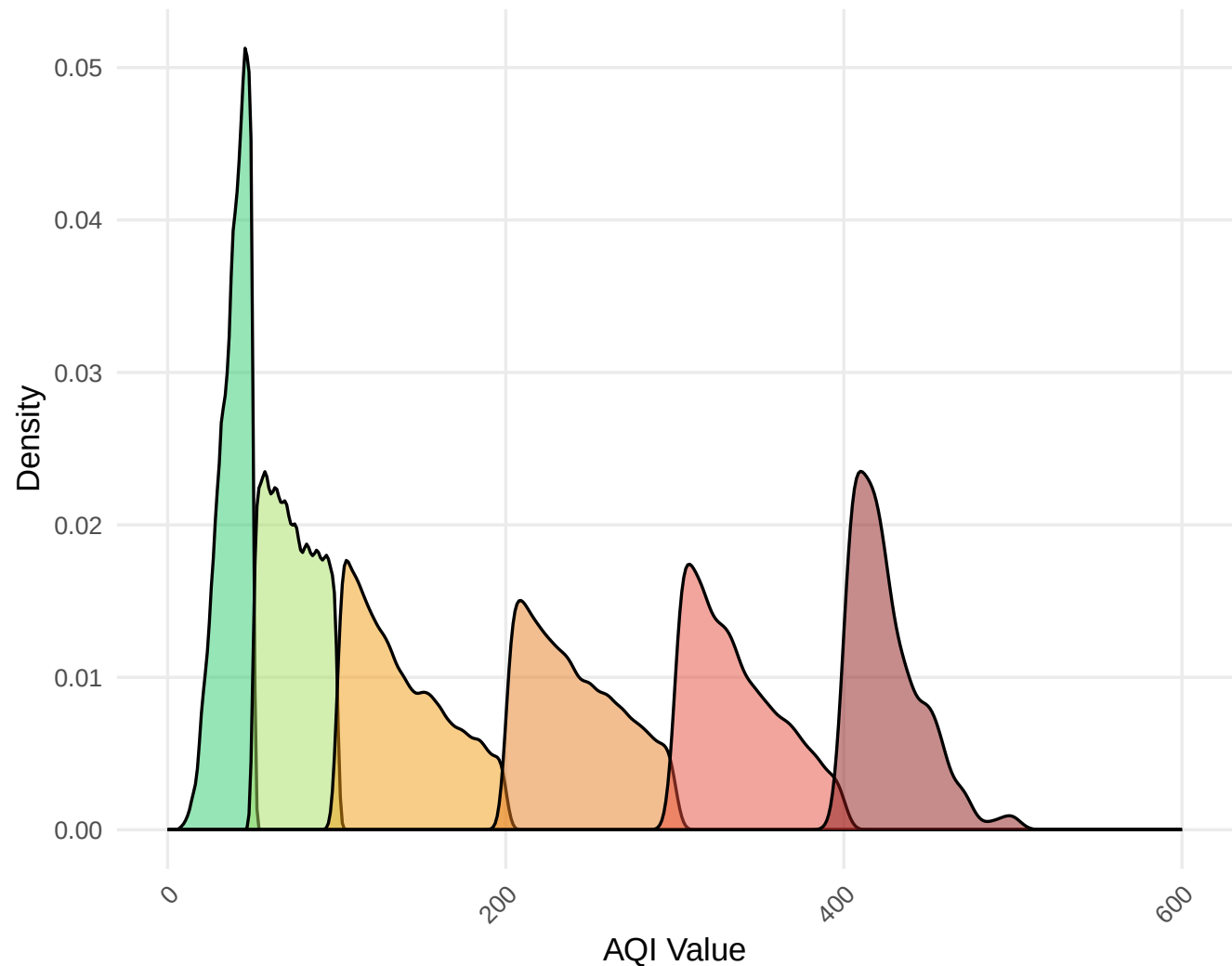
AQI Distribution by Top 10 Polluted States

Box plot showing median, IQR, and outliers



AQI Density Distribution by Air Quality Status

Overlapping density curves per category



Air Quality



Good



Moderate



Very Poor



Satisfactory



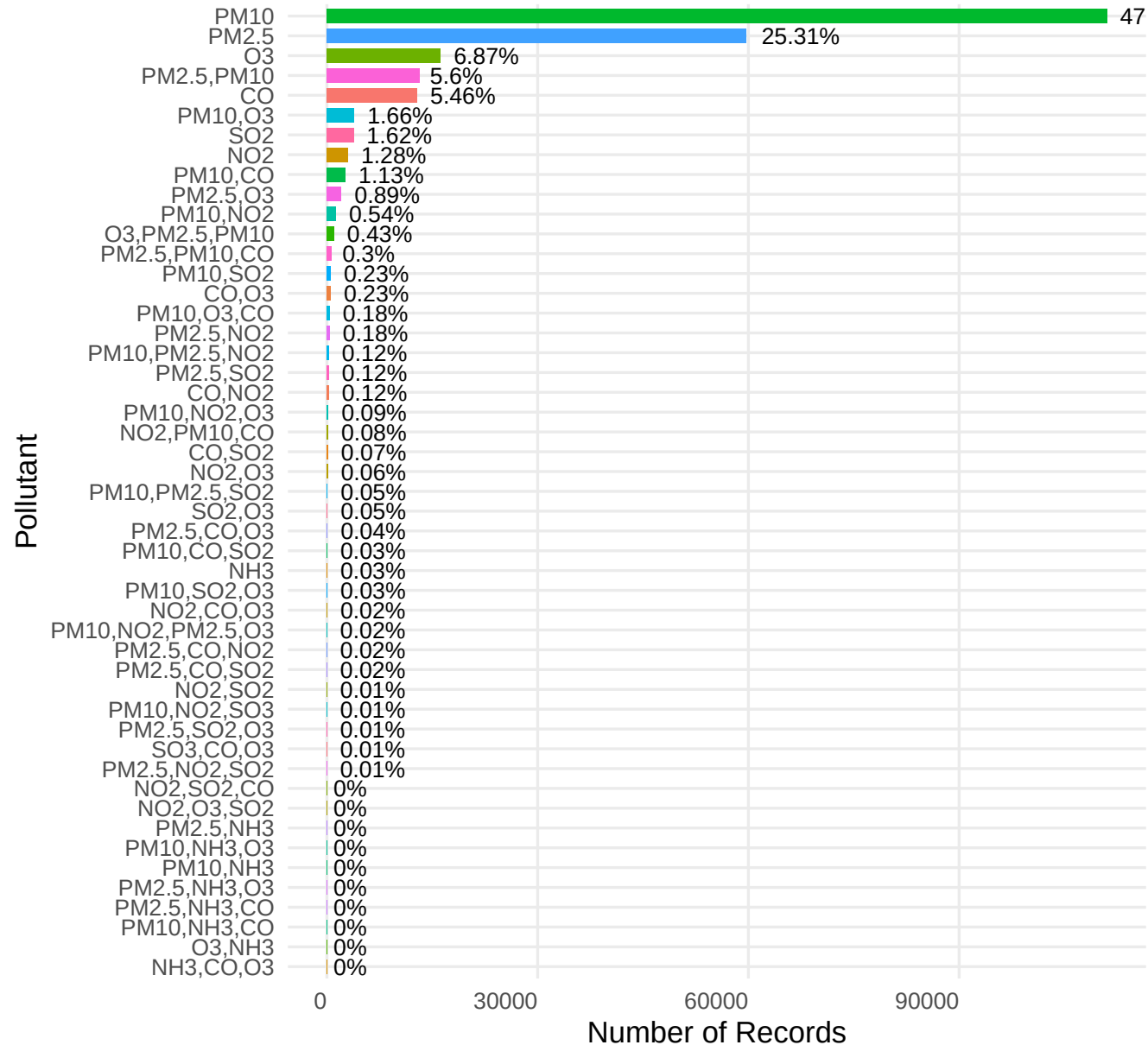
Poor



Severe

Distribution of Prominent Pollutants

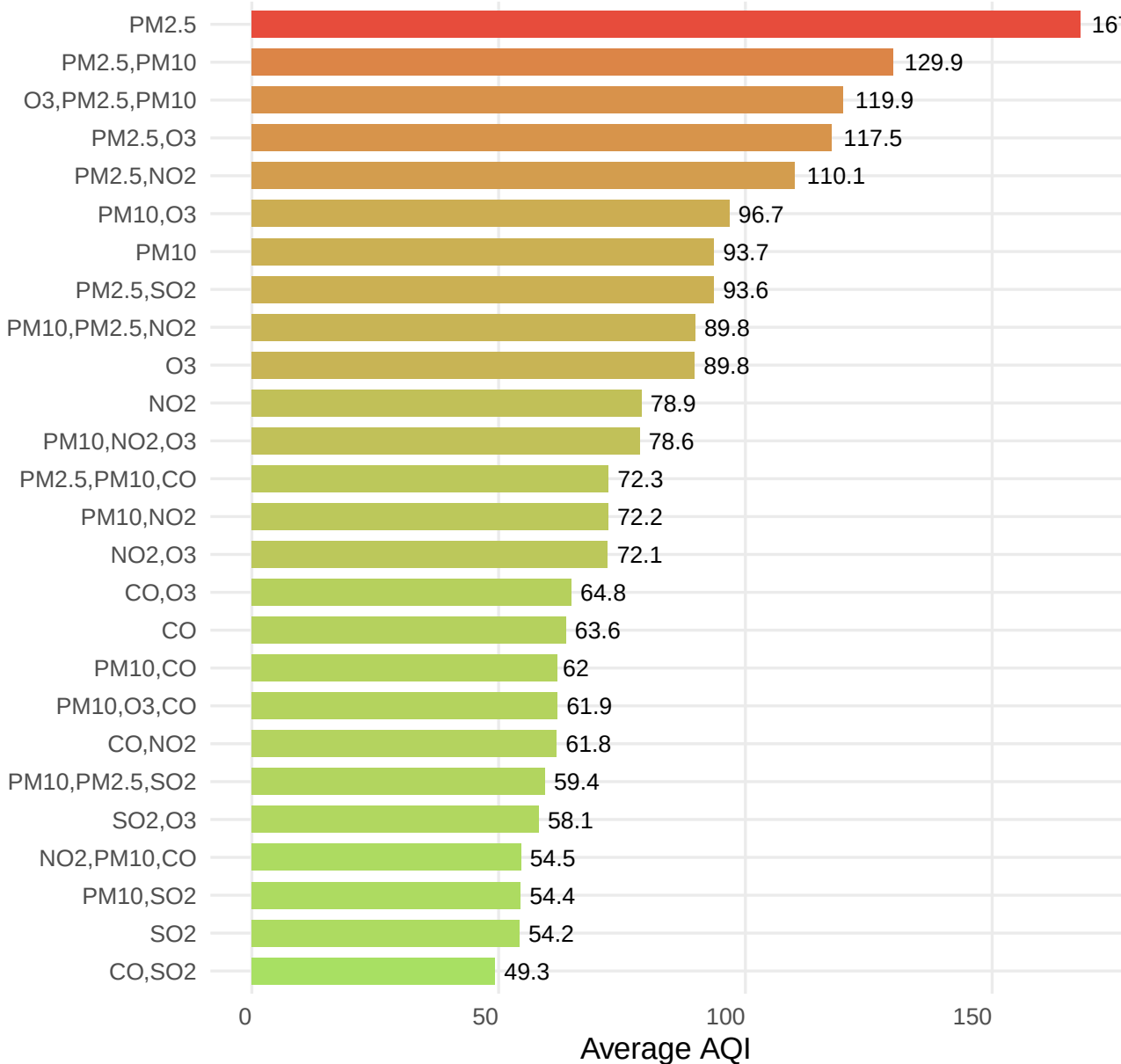
Which pollutant is most frequently the primary cause of pollution?



Average AQI by Prominent Pollutant

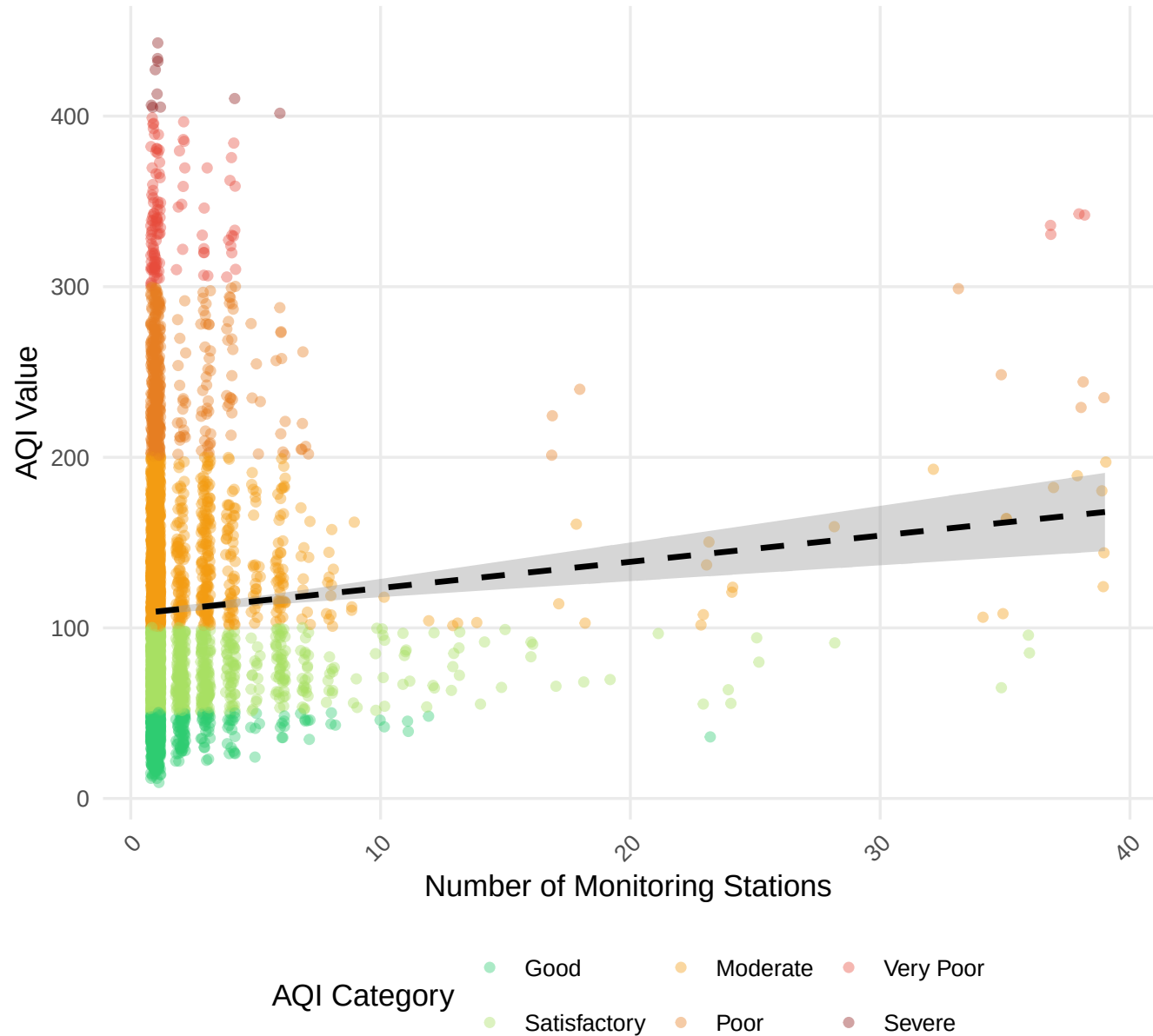
Which pollutant is associated with highest AQI?

Pollutant



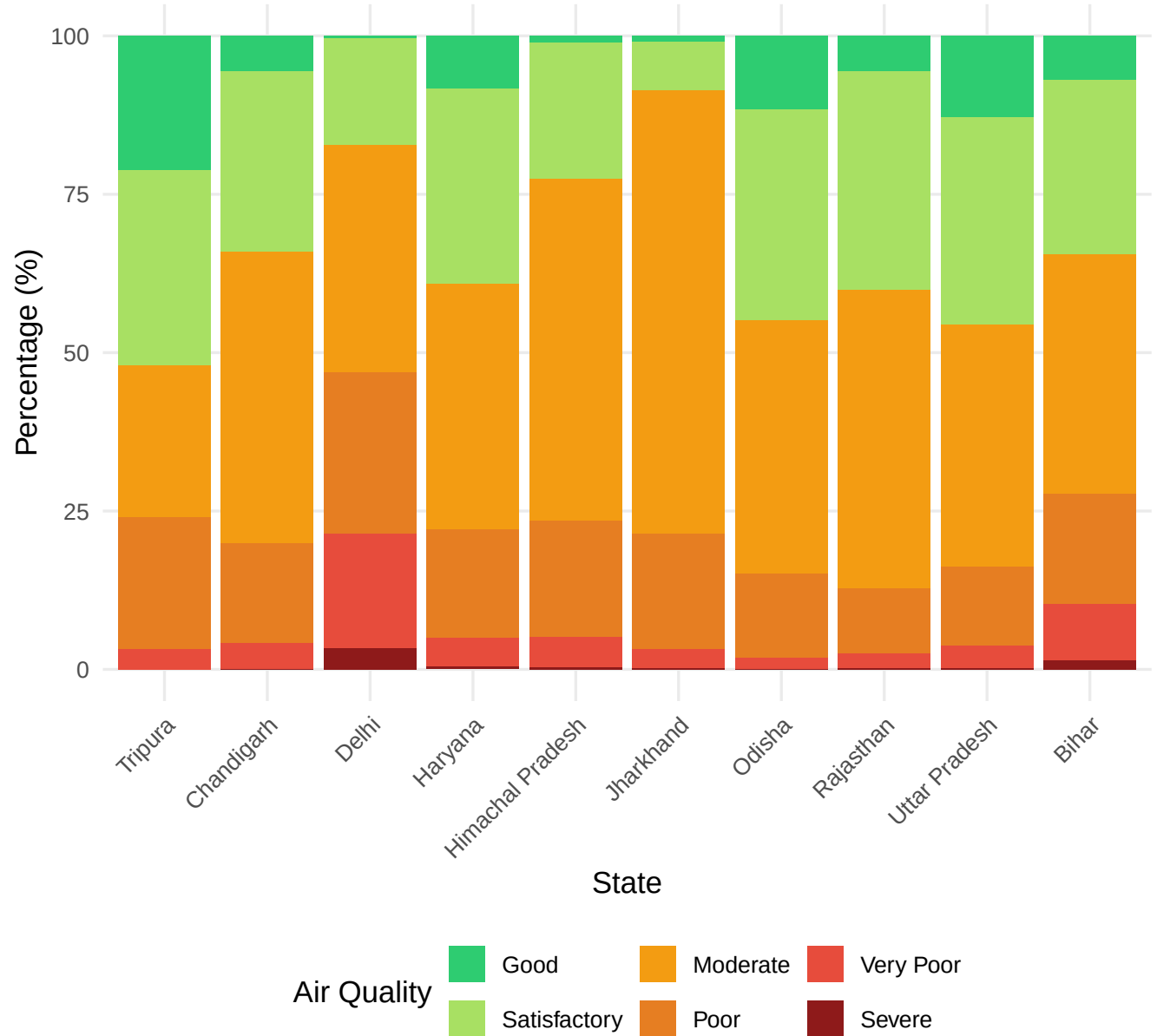
Monitoring Stations vs AQI Value

Each point is one observation (sampled 5000 rows)



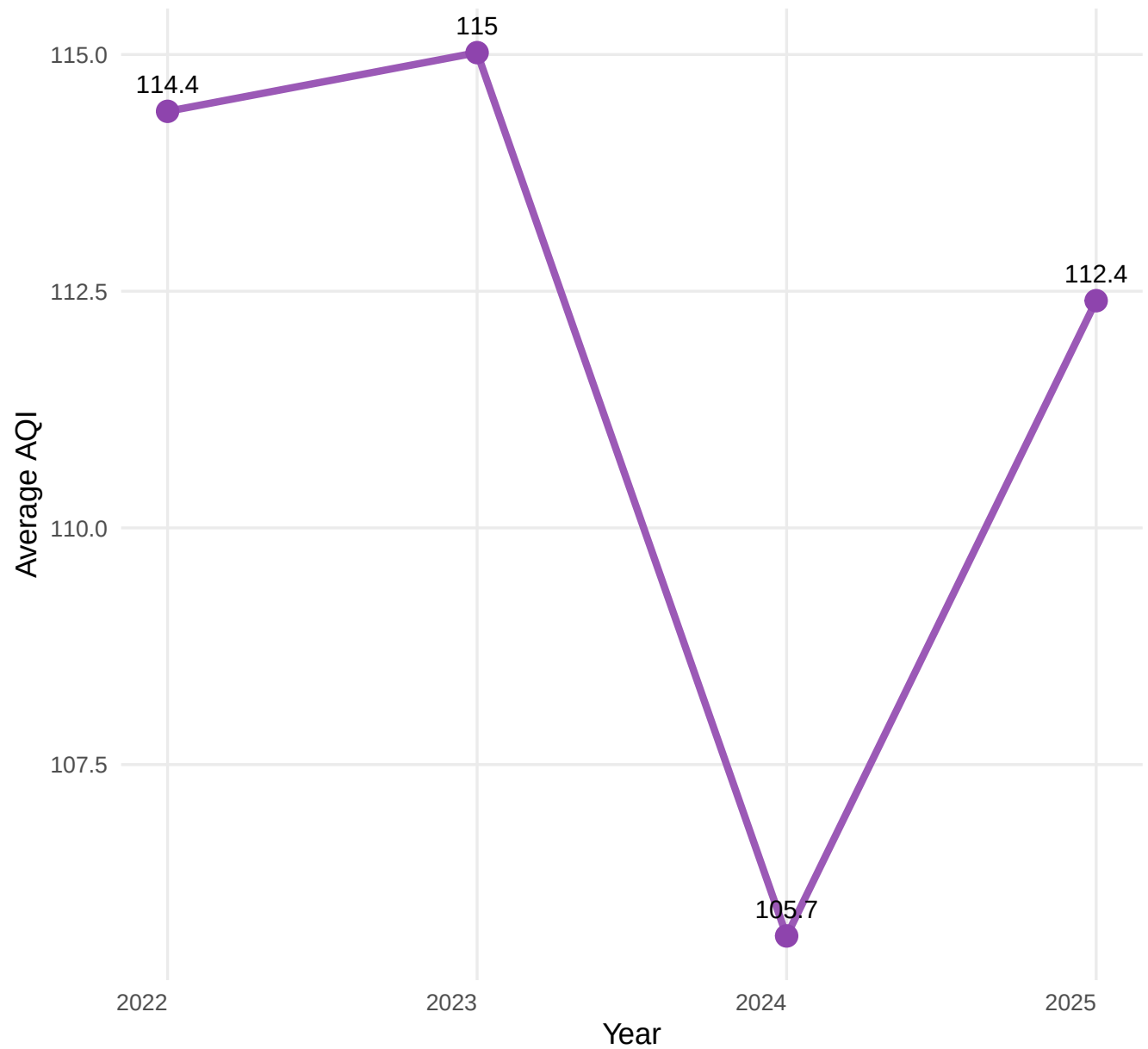
AQI Category Composition by State

Top 10 most polluted states



Year-wise AQI Trend

Average AQI across all India per year



Average AQI Heatmap: Top States by Month

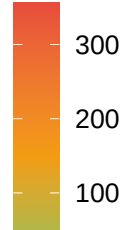
Darker red = worse air quality

State

Uttar Pradesh	160	123	109	138	136	123	63	70	78	144	201	166
Tripura	219	200	150	126	91	69	47	65	63	107	150	222
Rajasthan	165	132	119	132	138	110	82	79	80	121	193	176
Odisha	194	145	138	129	99	92	58	66	61	91	154	168
Jharkhand	194	158	151	172	153	187	138	153	121	160	230	163
Himachal Pradesh	263	160	132	154	175	167	85	96	108	169	203	224
Haryana	182	140	112	150	156	133	75	75	84	153	239	186
Delhi	324	223	172	207	202	166	89	94	106	221	356	320
Chandigarh	264	161	114	124	149	130	64	68	82	130	208	213
Bihar	246	188	147	161	132	123	79	80	79	120	230	252
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec

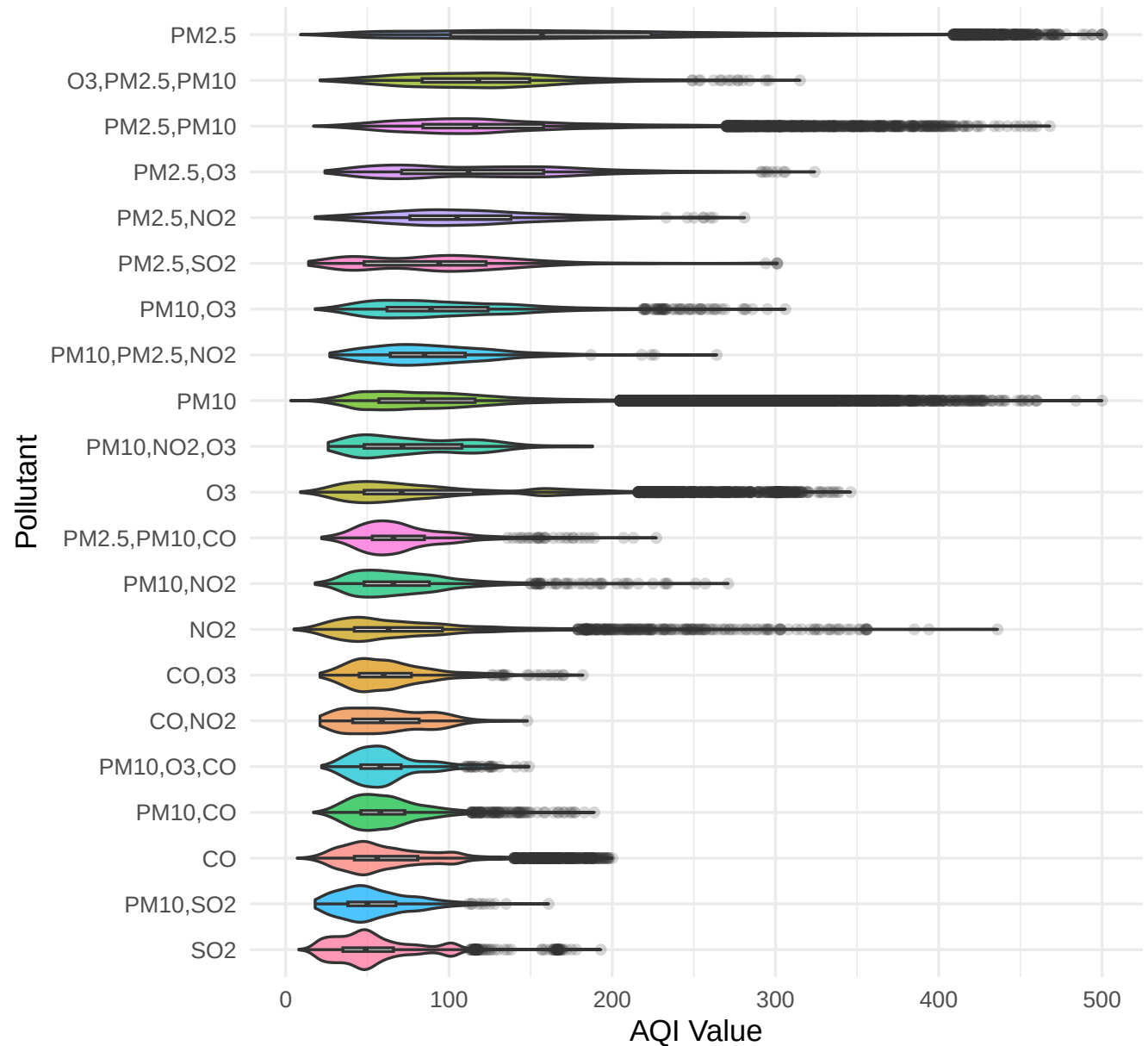
Month

Avg AQI



AQI Distribution by Pollutant Type (Violin + Box)

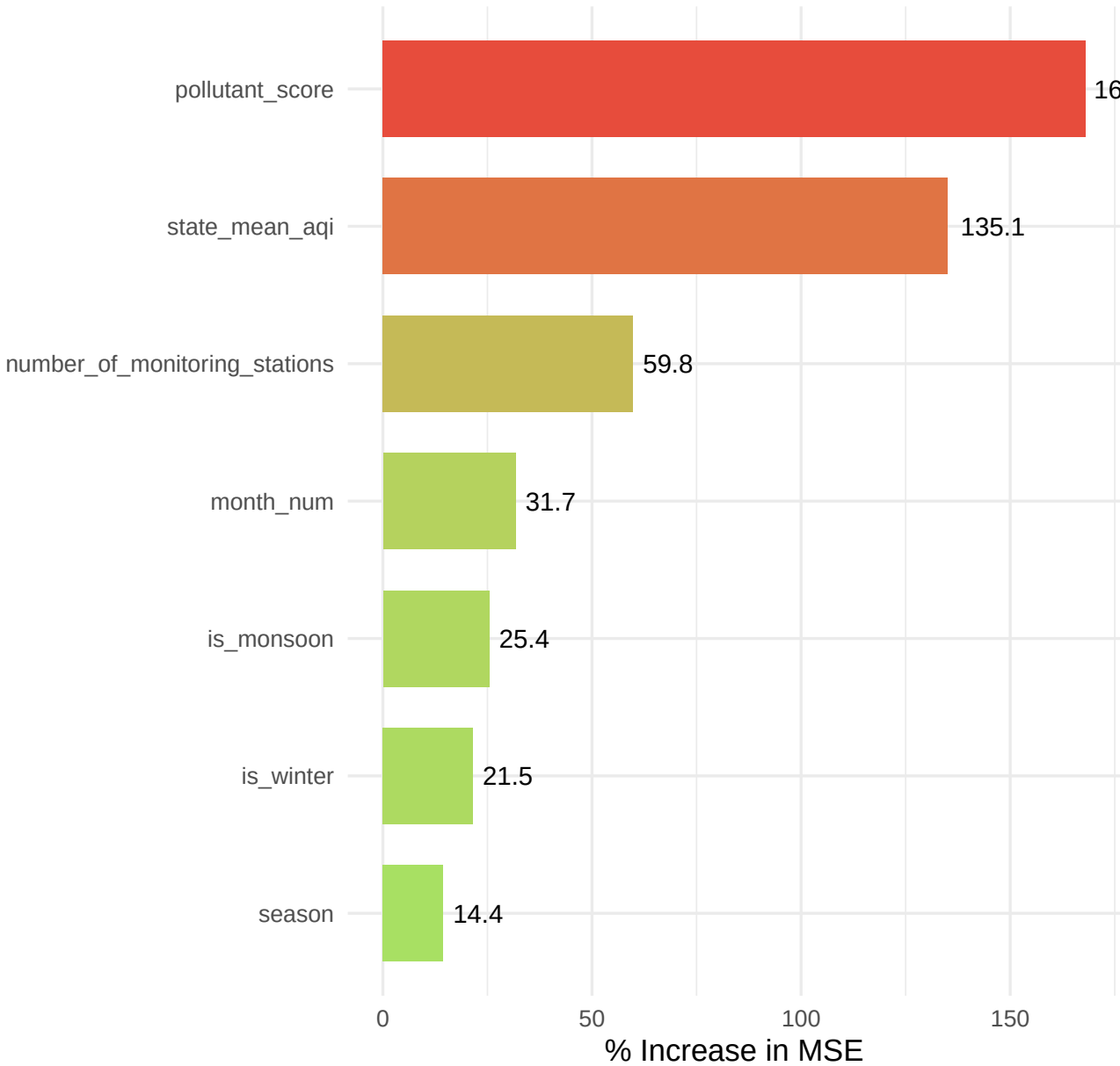
Statistical spread of AQI for each pollutant



Random Forest — Variable Importance

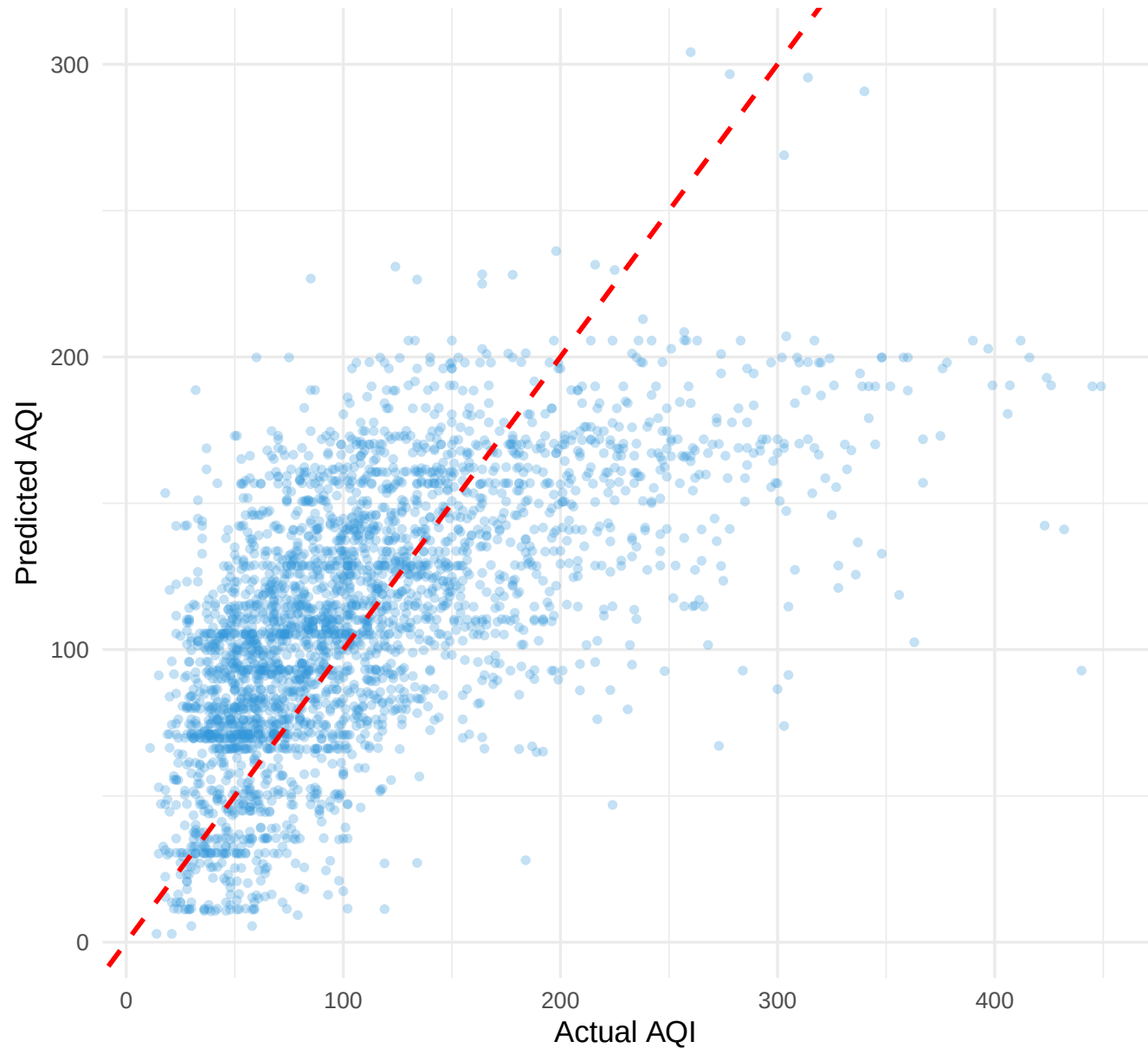
% Increase in MSE when feature is permuted

Feature



Linear Regression: Actual vs Predicted AQI

RMSE: 55.61 | R^2 : 0.39



Random Forest: Actual vs Predicted AQI

RMSE: 48.29 | R^2 : 0.541

